

Seminar 4.
Problem 1.

$$a) \lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \dots + \frac{1}{n(n+1)(n+2)} \right)$$

$\swarrow$
 $\frac{1}{k(k+1)(k+2)}$

$$\sum_{k=1}^n \frac{1}{k(k+1)(k+2)}$$

$$\frac{1}{k(k+1)(k+2)} = \frac{A}{k} + \frac{B}{k+1} + \frac{C}{k+2} =$$

$$= \frac{A(k^2+3k+2) + B(k^2+2k) + C(k^2+k)}{k(k+1)(k+2)}$$

$$k^2(A+B+C) + k(3A+2B+C) + 2A = 1 + 0k + 0k^2$$

$$\begin{cases} A+B+C = 0 \\ 3A+2B+C = 0 \\ 2A = 1 \end{cases} \Rightarrow A = \frac{1}{2}$$

$$\begin{cases} B+C = -\frac{1}{2} & (1) \\ 2B+C = -\frac{3}{2} & (2) \end{cases}$$

$$(1) - (2) : -B = 1 \Rightarrow B = -1 \Rightarrow C = \frac{1}{2}$$

$$\frac{1}{2} \cdot \frac{1}{k} - \frac{1}{k+1} + \frac{1}{2} \cdot \frac{1}{k+2}$$

$$\begin{array}{lcl} k=1 & : & \left(\frac{1}{2} \cdot \frac{1}{1} - \frac{1}{1+1} + \frac{1}{2} \cdot \frac{1}{1+2} \right) + \\ + & & \\ k=2 & : & \left(\frac{1}{2} \cdot \frac{1}{2} - \frac{1}{2+1} + \frac{1}{2} \cdot \frac{1}{2+2} \right) + \\ + & & \\ k=3 & : & \left(\frac{1}{2} \cdot \frac{1}{3} - \frac{1}{3+1} + \frac{1}{2} \cdot \frac{1}{3+2} \right) + \\ + & & \\ \dots & & \dots \\ k=n & : & \left(\frac{1}{2} \cdot \frac{1}{n} - \frac{1}{n+1} + \frac{1}{2} \cdot \frac{1}{n+2} \right) + \end{array}$$

$$\frac{1}{2} \left(\underbrace{1 + \frac{1}{2} + \cancel{\frac{1}{3}} + \cancel{\frac{1}{4}} + \dots + \frac{1}{n}}_{1er} + \underbrace{\cancel{\frac{1}{3}} + \cancel{\frac{1}{4}} + \frac{1}{5} + \dots + \frac{1}{n+2}}_{3er} \right) -$$

$$= \left(\cancel{\frac{1}{2}} + \cancel{\frac{1}{3}} + \cancel{\frac{1}{4}} + \dots + \frac{1}{n+1} \right)$$

$$S_n = \frac{1}{2} \left(1 + \frac{1}{2} + \frac{1}{n+2} \right) - \frac{1}{2} - \frac{1}{2(n+1)} =$$

$$S_n = \frac{1}{4} - \frac{1}{2(n+2)(n+1)}$$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{4} - \frac{1}{2(n+2)(n+1)} \right) = \frac{1}{4}$$

Problem 3.

and remarkable limit:
 $\{1^\infty\}$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

n - polynomial

$$\lim_{n \rightarrow \infty} \left(1 + \frac{a}{f(x)} \right)^{\frac{f(x)}{a}} = e$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{a}{f(x)} \right)^{q(x)} = \left[q(x) = \frac{f(x)}{a} \cdot p \right] = e^p$$

$$\begin{aligned} \text{a) } \lim_{n \rightarrow \infty} \left(\frac{n^2 + 2022}{n^2 + 2024} \right)^{n^2 + 2026} &= \left(\frac{\infty^2 + 2022}{\infty^2 + 2024} \right)^{\frac{\infty^2 + 2026}{\infty}} \\ &= \frac{n^2 + 2022}{n^2 + 2024} = 1 - \frac{2}{n^2 + 2024} \\ &= \frac{1 + \frac{2022}{n^2}}{1 + \frac{2024}{n^2}} = \{1^\infty\} \end{aligned}$$

$$= \lim_{n \rightarrow \infty} \left(1 + \frac{(-2)}{n^2 + 2024} \right)^{n^2 + 2026} = \left[n^2 + 2026 = \left[\frac{n^2 + 2024}{-2} \right] \cdot (-2) + 2 \right]$$

$$\begin{aligned}
 &= \lim_{n \rightarrow \infty} \left(1 + \frac{(-2)}{n^2+2024} \right)^{\left[\frac{n^2+2024}{-2} \right] \cdot (-2) + 2} \quad a^{b+c} = a^b \cdot a^c \\
 &= \lim_{n \rightarrow \infty} \underbrace{\left(1 + \frac{(-2)}{n^2+2024} \right)^{\frac{n^2+2024}{-2} \cdot (-2)}}_{e^{-2}} \cdot \lim_{n \rightarrow \infty} \underbrace{\left(1 + \frac{(-2)}{n^2+2024} \right)^2}_{1} =
 \end{aligned}$$

$\{1+0\}^2 = 1$

$$= e^{-2} = \frac{1}{e^2}$$

$$\begin{aligned}
 &\lim_{n \rightarrow \infty} \left(1 + \frac{a}{f(x)} \right)^{q(x)} = \\
 &\quad \left[q(x) = \left[\frac{f(x)}{a} \cdot b + c \right] \right] \\
 &= \lim_{n \rightarrow \infty} \underbrace{\left(1 + \frac{a}{f(x)} \right)^{\frac{f(x)}{a} \cdot b}}_{e^b} \cdot \underbrace{\left(1 + \frac{a}{f(x)} \right)^c}_{1^c = 1} = e^b
 \end{aligned}$$

b) $\lim_{n \rightarrow \infty} (8n+5) \left(\ln(4n-2) - \ln(4n+1) \right) =$

$a \cdot \log_e b = \log_e b^a$

$$\begin{aligned}
 &= \lim_{n \rightarrow \infty} (8n+5) \cdot \ln \left(\frac{4n-2}{4n+1} \right) = \lim_{n \rightarrow \infty} \ln \left(\frac{4n-2}{4n+1} \right)^{8n+5} \\
 &\quad \lim_{n \rightarrow \infty} \left(\frac{4n-2}{4n+1} \right)^{8n+5} = \{1^\infty\} = \\
 &= \left[\frac{4n+1-3}{4n+1} = 1 + \frac{(-3)}{4n+1} \right] = \\
 &= \lim_{n \rightarrow \infty} \left(1 + \frac{(-3)}{4n+1} \right)^{8n+5} = \left[8n+5 = \left[\frac{4n+1}{-3} \right] \cdot (-3) \cdot 2 + 3 \right] \\
 &= \lim_{n \rightarrow \infty} \underbrace{\left(1 + \frac{(-3)}{4n+1} \right)^{\left[\frac{4n+1}{(-3)} \right] \cdot (-3) \cdot 2 + 3}}_{e^{-3 \cdot 2} \cdot 1^3 = e^{-6}}
 \end{aligned}$$

$$\ln e^{-6} = -6$$

$$\ln = \log_e$$

Problem 5.

$$a) \lim_{n \rightarrow \infty} \frac{n}{n^3+1} \cdot \cos(n+7)$$

$$\begin{aligned} -1 \leq \cos(n+7) \leq 1 & \quad \Big| \cdot \frac{n}{n^3+1} \\ -\frac{n}{n^3+1} \leq \frac{n \cos(n+7)}{n^3+1} \leq \frac{n}{n^3+1} \end{aligned}$$

$\downarrow$ c_n $\downarrow$ a_n $\leftarrow$ b_n

$$c_n \leq a_n \leq b_n, \quad \text{if} \quad \lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} b_n = L \Rightarrow \lim_{n \rightarrow \infty} a_n = L.$$

$$\lim_{n \rightarrow \infty} \frac{-n}{n^3+1} = \left\{ \frac{\infty}{\infty} \right\} = \lim_{n \rightarrow \infty} \frac{\frac{-1}{n^2} \rightarrow 0}{1 + \frac{1}{n^3} \rightarrow 0} = \frac{0}{1} = 0.$$

$$\lim_{n \rightarrow \infty} \frac{n}{n^3+1} = \left\{ \frac{\infty}{\infty} \right\} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n^2} \rightarrow 0}{1 + \frac{1}{n^3} \rightarrow 0} = \frac{0}{1} = 0$$

$$c) \lim_{n \rightarrow \infty} \frac{(3n-7)^4 \left((2n + \sqrt{\sin n} - 1) \right)^2}{(5n^2 + n - 2)^3}$$

$$-1 \leq \sin n \leq 1 \quad \Big| + (2n-1)$$

$$2n-2 \leq 2n + \sin n - 1 \leq 2n$$

$$(2n-2)^2 \leq (2n + \sin n - 1)^2 \leq (2n)^2 \quad \Big| \frac{(3n-7)^4}{(5n^2+n-2)^3}$$

$$\underbrace{\frac{(2n-2)^2 (3n-7)^4}{(5n^2+n-2)^3}}_{c_n} \leq \underbrace{\hspace{10em}} \leq \underbrace{\frac{(2n)^2 (3n-7)^4}{(5n^2+n-2)^3}}_{b_n}$$

the way: $\cdot (5n^2+n-2)^3 = 125n^6 + \underbrace{\dots}_{\text{power } n < 6} + -8$

$$\begin{aligned}
 & \bullet (3n-7)^4 = 81n^4 + \overbrace{\dots (-7)^4}^{\text{power } n < 4} \\
 & \bullet (2n + \sin n - 1)^2 = 4n^2 + \underbrace{\dots}_{-1 \leq \sin n \leq 1} + 1 \quad \text{power } n < 2 \\
 & \qquad \qquad \qquad \frac{324n^6 + \dots + 1}{125n^6 + \dots - 8} = \frac{324}{125}
 \end{aligned}$$

Problem 6.

convergent = monotonicity + bounded

$$\begin{aligned}
 & \downarrow \\
 & \bullet x_{n+1} > x_n \\
 & \bullet x_{n+1} < x_n \\
 & \downarrow \\
 & x_{n+1}^2 - x_n^2 = 2x_n - x_n^2 = x_n(2-x_n) \\
 & \quad x_n > 0 \quad \text{and} \quad 2-x_n > 0
 \end{aligned}
 \qquad \begin{aligned}
 & x_1 = \sqrt{2}. \\
 & \forall n \in \mathbb{N} \\
 & x_n > 0.
 \end{aligned}$$

$$\Rightarrow x_{n+1} > x_n.$$

The sequence is increasing and bounded $\Rightarrow$
 By Weierstrass th it has limit.

$$\lim_{n \rightarrow \infty} x_n = A.$$

$x_{n+1} = \sqrt{2x_n}$; the left side tends to A , the right side to $\sqrt{2A}$

$$\begin{aligned}
 \Rightarrow \quad & A = \sqrt{2A} \\
 & A^2 - 2A = 0 \\
 & A(A-2) = 0 \\
 & A = 0 \quad \text{or} \quad A = 2.
 \end{aligned}$$

as by bound, $x_n > 0 \Rightarrow \lim = 2.$

Task 7a.

find the limit of this sequence if (a) $x_1 = 1$; (b) $x_1 = 7$.

Solution

(a)

Prove that $\{x_n\}$ is bounded. We will prove that $x_n < 5$ by induction on n . Notice that $x_n \geq 0$ for any $n \in \mathbb{N}$.

Base case. $x_1 = 1 < 5$. Inductive step. Assume that $x_n < 5$. Then $x_{n+1} = \sqrt{15 + 2x_n} < \sqrt{15 + 10} = 5$.

Prove that $\{x_n\}$ is monotone increasing. Find the difference:

$$x_{n+1}^2 - x_n^2 = 15 + 2x_n - x_n^2 = (5 - x_n)(x_n + 3).$$

Since $5 - x_n > 0$ and $x_n + 3 > 0$, we see that $x_{n+1}^2 - x_n^2 > 0$ as required.

Use Weierstrass Theorem. Since this sequence is monotone increasing and bounded above, we can conclude that it is convergent.

Find the limit. Since $\{x_n\}$ is convergent and $x_{n+1} = \sqrt{15 + 2x_n}$, we can consider limits of this equality and use arithmetic properties of limits to conclude that

$$\lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} \sqrt{15 + 2x_n} = \sqrt{15 + 2 \lim_{n \rightarrow \infty} x_n}.$$

Denote $\lim_{n \rightarrow \infty} x_n$ by A and come to equation

$$A = \sqrt{15 + 2A} \iff A^2 - 2A - 15 = 0 \iff A = 5 \text{ or } A = -3.$$

Since $x_n \geq 0$, the limit cannot be equal to -3 , so it is equal to 5 .

Task 8.

boundedness: $x_1 = \frac{1}{2}$
 $n \in \mathbb{N} \Rightarrow n > 0$

$$x_{n+1} = \frac{4}{3}x_n - x_n^2, \quad x_1 = \frac{1}{2}$$

monot: $x_{n+1} - x_n > 0$
 $\frac{4}{3}x_n - x_n^2 - x_n > 0$
 $\frac{1}{3}x_n - x_n^2 > 0$
 $x_n \left(\frac{1}{3} - x_n \right) > 0 \quad \downarrow \text{false}$

Try w/w $\Rightarrow x_{n+1} - x_n < 0$
 $x_n \left(\frac{1}{3} - x_n \right) < 0 \quad \text{true}$

decreasing

lim: $A = \frac{4}{3}A - A^2$
 $\Rightarrow A = 0 \quad \text{or} \quad A = \frac{1}{3}$

$$x_n > 0 \Rightarrow \lim_{n \rightarrow \infty} x_n = \bar{c}$$